1. Ellingham diagram for the formation of Ag_2O and HgO is at upper part of the diagram and their decomposition temperatures are 600 and 700 K respectively. It indicates that these oxides are unstable at moderate temperatures and will decompose on heating even in the absence of a reducing agent.
2. Ellingham diagram is used to predict thermodynamic feasibility of reduction of oxides of one metal by another metal. Any metal can reduce the oxides of other metals that are located above it in the diagram. For example, in the Ellignham diagram, for the formation of chromium oxide lies above that of the aluminium, meaning that Al_2O_3 is more stable than Cr_2O_3. Hence aluminium can be used as a reducing agent for the reduction of chromic oxide. However, it cannot be used to reduce the oxides of magnesium and calcium which occupy lower position than aluminium oxide.
3. The carbon line cuts across the lines of many metal oxides and hence it can reduce all those metal oxides at sufficiently high temperature. Let us analyse the thermodynamically favourable conditions for the reduction of iron oxide by carbon. Ellingham diagram for the formation of FeO and CO intersects around 1000 K. Below this temperature the carbon line lies above the iron line which indicates that FeO is more stable than CO and hence at this temperature range, the reduction is not thermodynamically feasible. However, above 1000 K carbon line lies below the iron line and hence, we can use coke as reducing agent above this temperature. The following free energy calculation also confirm that the reduction is thermodynamically favoured.

   From the Ellingham Diagram at 1500 K,

$$2Fe\ (s) + O_2\ (g) \longrightarrow 2FeO\ (g) \qquad \Delta G_1 = -350\ kJ\ mol^{-1}\ \text{------}\ (1)$$

$$2C\ (s) + O_2\ (g) \longrightarrow 2CO\ (g) \qquad \Delta G_2 = -480\ kJ\ mol^{-1}\text{------}\ (2)$$

   Reverse the reaction (1)

$$2FeO\ (s) \longrightarrow 2Fe\ (s) + O_2\ (g) \quad -\Delta G_1 = +350\ kJ\ mol^{-1}\ \text{------}\ (3)$$

   Now couple the reactions (2) and (3)

$$2FeO\ (s) + 2C \longrightarrow 2Fe\ (l,s) + 2CO\ (g) \quad \Delta G_3 = -130\ kJ\ mol^{-1}\ \text{------}\ (4)$$

The standard free energy change for the reduction of one mole of FeO is, $\Delta G_3/2 = -65\ kJ\ mol^{-1}$

Limitations of Ellingham diagram

1. Ellingham diagram is constructed based only on thermodynamic considerations. It gives information about the thermodynamic feasibility of a reaction. It does not tell anything about the rate of the reaction. More over, it does not give any idea about the possibility of other reactions that might be taking place.
2. The interpretation of ΔG is based on the assumption that the reactants are in equilibrium with the products which is not always true.

XII U1 Metallurgy - Jerald.indd 13 04-01-2021 12:41:35